USDA Plant Diseases and Pests Manager

Developer: Aneeb Zulfiqar
Stack: Django (Backend) | Next.js (Frontend) | SQLite (Database)

Summary

The **Plant Diseases and Pests Manager** is a full-stack concept designed to streamline the reporting, tracking, and moderation of invasive pests and plant diseases listed on the USDA APHIS website. The web application features a dashboard for visualizing the data and a secure administrative back-end for data integrity and validation of entries.

Technical Architecture

- **Backend:** Django REST Framework (DRF) manages the API, providing security and structured JSON data to the frontend.
- **Frontend:** Next.js provides a responsive UI with server-side or client side rendering for efficient data fetching and display.
- **Admin Control:** Utilizes Django's built-in Admin Panel to allow administrators to approve "Pending" entries or "Removal Requests" before they may affect a live public database.

Database Schema

The system is relatively simple currently. It uses a relational database structure to ensure data normalization and hierarchical categorization and also ability to scale in the future.

Entities (Tables):

- **Pest/Disease Table:** Stores scientific/common names, descriptions, and treatment protocols.
- **Category Table:** Uses a self referencing `Parent_ID` to allow for nested hierarchies (e.g., *Insects* nested under *Pests*).
- **Status Workflow:** Every entry contains a `Status` (ACTIVE, PENDING, or REMOVAL_REQUESTED) to simulate a government review process.

Key Features

- **Dynamic Filter Search:** Quick filtering of the database by status (Active, Pending, Removal Requested)
- **Request Pipeline:** Users can submit "Addition" or "Deletion" requests via structured forms.
- **Identification Guide:** Detailed views include "What to look for" and "Prevention" fields, mimicking real-world APHIS outreach materials (same as the APHIS website section).

Entity Examples

Table 1 - Category Table

Field	Description
Name	The label for the group
Parent ID	Links a sub group to a main group (Ants → Insects or Plum Pox → Diseases)

EXAMPLE:

ID	Name	Parent (Parent_ID)
1	Pests	-
2	Diseases	-
3	Insects and Mites	Pests (1)
4	Citrus Diseases	Diseases (2)

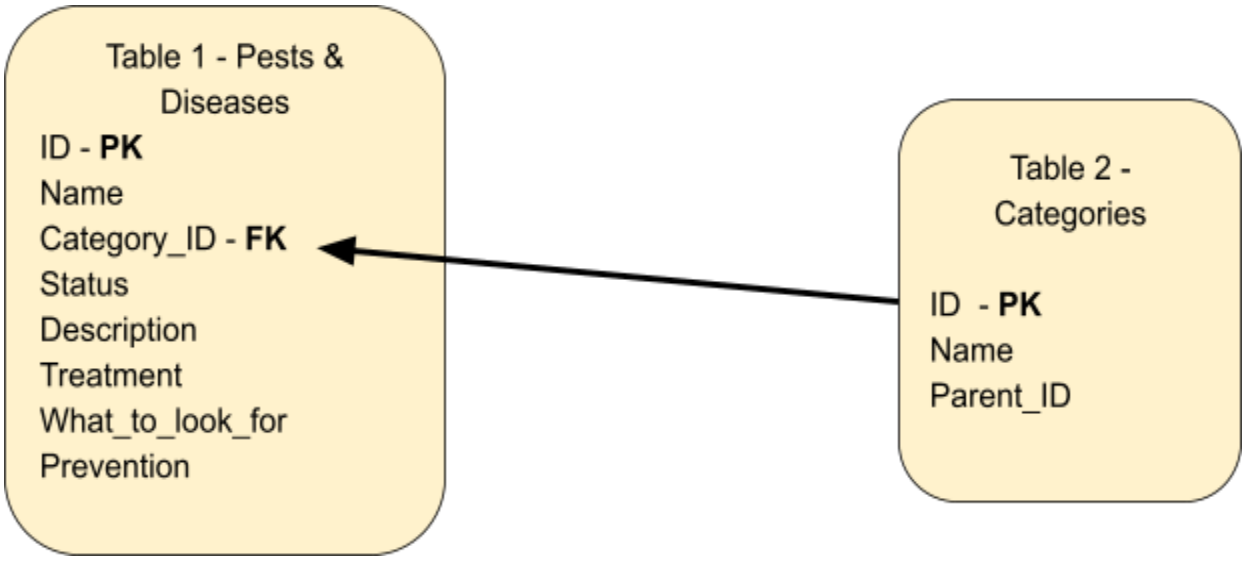
Table 2 - Pest and Diseases Table

Field Name	Type	Purpose
Common Name	String	Common Name
Category ID	Foreign Key	Links this item to a specific Category.
Status	Choice	PENDING, ACTIVE, or REMOVAL
Description	Text	The general blurb about the pest.
What to look for	Text	Identification signs (mounds, leaf spots).
Treatment	Text	How APHIS or owners treat the issue.
Prevention	Text	Quarantine and regulatory info.

EXAMPLE:

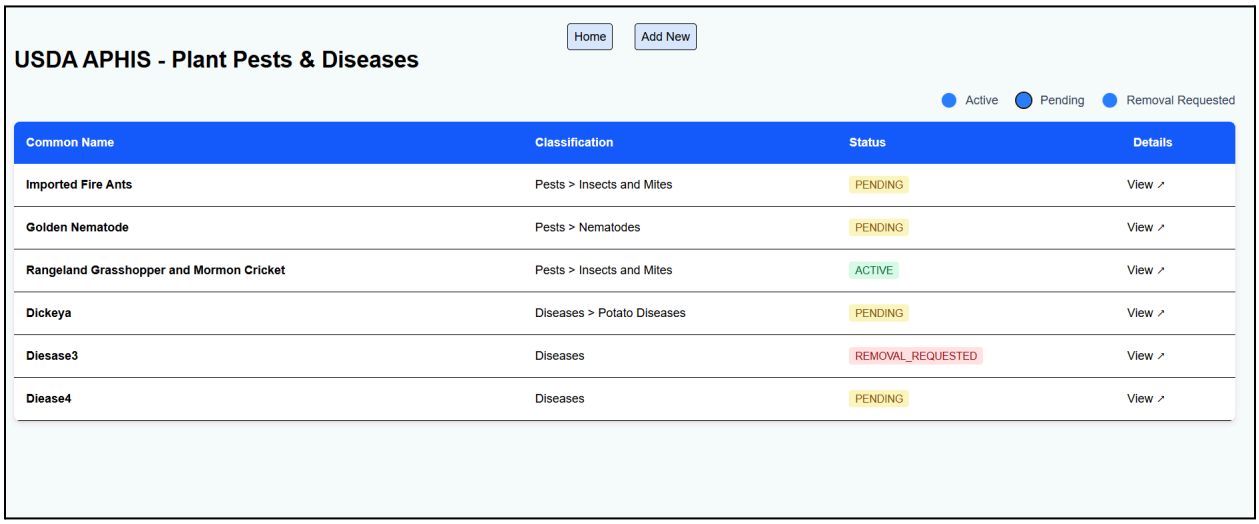
ID	Name	CatID	Status	Description	Treatment…
1	Imported Fire Ants	3	Active	Desc……..	…
2	Plum Pox	2	Pending	Desc…….	…
3	Honey Bee	3	Removal Requested	Desc……	…
4	Citrus Black Spot	4	Active	Desc…..	…

<u>Entity Relationship Diagram (ERD)</u>

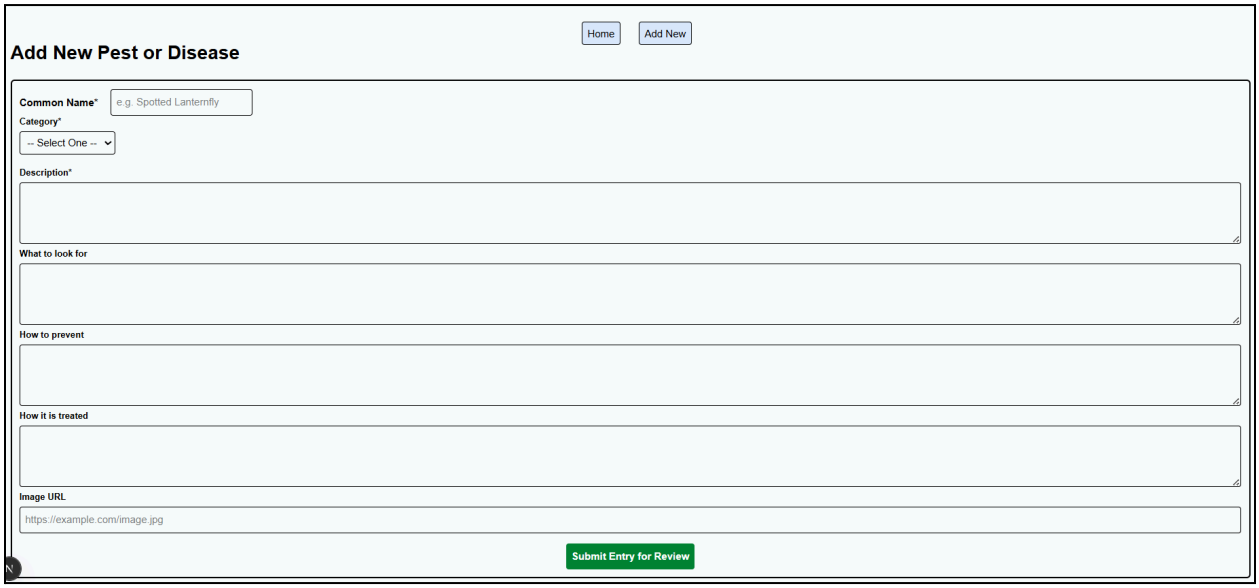


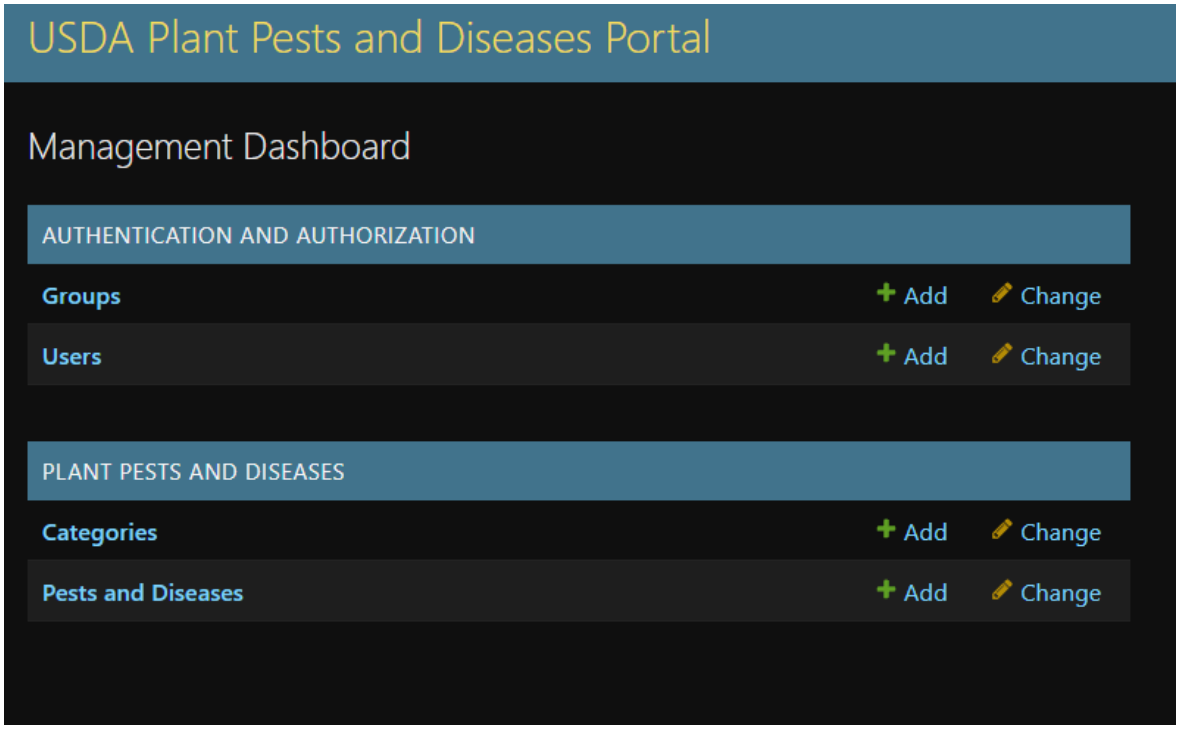
User Interface Screenshots

Main Dashboard Page



Add New Page



Details and Removal Request Page

Admin Panel